DYSLEXIA AWARE SEMANTIC GRADER SYSTEM

A
Industrial Oriented Mini Project

Submitted in partial fulfillment of
the requirements for the award of the degree of

Bachelor of Technology
in
Computer Science and Engineering(AIandML)

Submitted by

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Certificate

This is to certify that the Mini Project Report work entitled **"DYSLEXIA AWARE SEMANTIC GRADER SYSTEM "** is a bonafide work carried out by a team consisting of **P. PavithraChary,** bearing Roll no.**23SS1A6648, B. Mounika ,** bearing Roll no.**23SS1A6608, P. Ram Charan** bearing Roll no.**23SS1A6649, P. Adith** bearing Roll no.**23SS1A6645,** in partial fulfillment of the requirements for the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING(AI and ML) discipline to Jawaharlal Nehru Technological University Hyderabad University College of Engineering Sultanpur during the academic year 2025-2026.

The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

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Declaration

We hereby declare that Industrial Oriented Mini Project entitled "**DYSLEXIA AWARE SEMANTIC GRADER** " is a bonafide work carried out by a team consisting of **P. Pavithra Chary** bearing Roll no.**23SS1A6648, B. Mounika** bearing Roll no.**23SS1A6608, P. Ram Charan** bearing Roll no.**23SS1A6649, P. Adith** bearing Roll no.**23SS1A6645,**in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering(AI and ML) discipline to Jawaharlal Nehru Technological University Hyderabad College of Engineering Sultanpur during the academic year 2024-2025. The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

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Abstract

Dyslexia is a learning disorder that affects reading, spelling, and writing skills, often leading to unfair evaluation in traditional grading systems that emphasize linguistic accuracy over conceptual understanding. To address this issue, this project proposes **DYSLEXIA-AI**, a **Dyslexia-Aware Semantic Grader** that evaluates student responses based on meaning rather than surface-level errors.The system leverages Artificial Intelligence and Natural Language Processing (NLP) techniques such as semantic similarity, keyword extraction, and context analysis to identify conceptual correctness even in the presence of spelling and grammatical mistakes. Deep Learning models like ResNet50 are used for handwriting recognition, enabling accurate interpretation of handwritten responses. Additionally, the system incorporates spell-tolerant mechanisms and flexible grammar evaluation to ensure fairness and inclusivity. It compares student answers with model responses using semantic embeddings and assigns scores based on content relevance and completeness, while also providing constructive feedback.This approach improves grading accuracy and can be applied in online assessments, educational platforms, and assistive learning systems to create a more inclusive evaluation environment.

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Chapter 1

Introduction

Dyslexia is a learning disorder that affects reading, spelling, and writing abilities. In educational settings, traditional grading systems focus heavily on linguistic accuracy, which often disadvantages students with dyslexia. A Dyslexia-Aware AI Semantic Grader evaluates answers based on the meaning Of the content rather than spelling mistakes.

In recent years, the use of automated grading systems has increased significantly in educational institutions due to their efficiency, consistency, and ability to handle large volumes of student responses. However, most traditional grading systems primarily focus on surface-level features such as spelling, grammar, and sentence structure, which can create challenges for students with learning difficulties like dyslexia. Dyslexia is a common learning disorder that affects reading, writing, and spelling skills, often making it difficult for students to express their knowledge accurately in written form. Conventional evaluation methods may unfairly penalize such students, even when they possess a clear understanding of the subject matter. This highlights the need for an inclusive assessment approach that emphasizes conceptual understanding rather than linguistic accuracy. To address this issue, the Dyslexia-Aware Semantic Grader is proposed as an intelligent system that evaluates answers based on their meaning rather than their form.

The system leverages Natural Language Processing (NLP) techniques to analyze and interpret the semantic content of student responses. By using methods such as semantic similarity, keyword matching, and context analysis, it can identify correct answers even if they contain spelling mistakes or unconventional phrasing. This ensures

that students are assessed fairly based on their knowledge and comprehension. Furthermore, the system incorporates spell-tolerant mechanisms and flexible grammar evaluation, making it more accessible and supportive for dyslexic learners. It also provides constructive feedback, helping students improve their communication skills without discouragement.

Overall, the Dyslexia-Aware Semantic Grader aims to promote inclusive education by reducing bias in evaluation and encouraging a more equitable learning environment. It represents a step forward in integrating artificial intelligence into education to support diverse learning needs and enhance academic assessment practices.

1.1 Project Overview

The Dyslexia-Aware Semantic Grader is an intelligent evaluation system designed to assess student answers based on their semantic meaning rather than strict adherence to spelling, grammar, and writing conventions. The primary goal of this project is to create a fair and inclusive grading mechanism that supports students with dyslexia and other language-related learning difficulties.

This system utilizes Natural Language Processing (NLP) techniques to analyze student responses and compare them with predefined model answers. Instead of relying solely on exact keyword matches, the grader interprets the context and intent behind the response using semantic similarity algorithms. This enables the system to recognize correct answers even when they include spelling errors, phonetic variations, or non-standard sentence structures.

The project is composed of several key components, including input processing, text normalization, semantic analysis, and scoring. The input processing stage cleans and prepares the student's response by removing unnecessary symbols and standardizing the text. In the semantic analysis stage, advanced techniques such as word embeddings and similarity measures are applied to evaluate how closely the student's answer matches the expected answer in terms of meaning. Additionally, the system incorporates a spell-tolerant mechanism to handle common dyslexic errors and ensures that minor linguistic mistakes do not negatively impact the score. The grading module then assigns marks based on the level of conceptual accuracy and completeness of the re-

sponse. Another important feature of the system is its feedback generation capability. Instead of simply providing a score, the system offers constructive and encouraging feedback, helping students understand their mistakes and improve their learning outcomes.

Overall, the Dyslexia-Aware Semantic Grader provides an efficient, scalable, and inclusive solution for modern educational environments. It can be integrated into online examination systems, learning management platforms, and educational applications to enhance the fairness and effectiveness of student assessment.

1.2 Problem Statement

Traditional automated grading systems primarily evaluate student responses based on correct spelling, grammar, and exact keyword matching. While these methods ensure consistency and speed, they often fail to account for students with learning disabilities such as dyslexia, who may struggle with writing accuracy despite having a clear understanding of the subject matter. Dyslexic learners frequently make spelling errors, reverse letters, or use unconventional sentence structures, which can lead to incorrect or unfair evaluation by conventional systems. As a result, these students may receive lower scores that do not accurately reflect their knowledge and comprehension, leading to reduced confidence and motivation.

Furthermore, existing grading systems lack the ability to interpret the semantic meaning or intent behind student responses. They depend heavily on rigid evaluation criteria, making them unsuitable for assessing descriptive or subjective answers where multiple correct expressions are possible. Therefore, there is a need for an intelligent and inclusive grading system that can evaluate answers based on their conceptual meaning rather than surface-level correctness. The problem addressed in this project is to design and develop a Dyslexia-Aware Semantic Grader that can understand, analyze, and evaluate student responses fairly by incorporating semantic analysis, spell-tolerant mechanisms, and flexible evaluation criteria.

The proposed system aims to bridge the gap between accurate knowledge assessment and inclusive education by ensuring that students are graded based on what they

know, not how perfectly they write.

1.3 Objective

The primary objective of the Dyslexia-Aware Semantic Grader is to develop an intelligent and inclusive system that evaluates student responses based on their semantic meaning rather than focusing solely on spelling, grammar, or writing style.

The specific objectives of this project are as follows:

1. To design a grading system that understands and evaluates the conceptual meaning of student answers using Natural Language Processing (NLP) techniques.

2. To implement semantic similarity methods that compare student responses with model answers and determine the level of correctness based on meaning.

3. To incorporate spell-tolerant mechanisms that can handle common dyslexic errors such as phonetic spelling, letter reversals, and minor misspellings.

4. To reduce bias in traditional grading systems by minimizing penalties for linguistic errors that do not affect the actual understanding of the subject.

5. To provide constructive and supportive feedback to students, helping them improve their learning without discouragement.

6. To develop a scalable and efficient system that can be integrated into online examination platforms and learning management systems.

7. To promote inclusive education by ensuring fair assessment for students with dyslexia and other language-related learning challenges.

Overall, the project aims to shift the focus of evaluation from writing accuracy to knowledge comprehension, thereby creating a more equitable and student-friendly assessment process.

1.4 Existing System

The existing automated grading systems used in educational environments primarily rely on rule-based or keyword-matching techniques to evaluate student responses. These systems are designed to compare student answers with predefined model answers by checking for exact word matches, correct spelling, and proper grammatical structure. Most traditional systems focus on surface-level features such as syntax, vocabulary, and formatting rather than understanding the actual meaning of the response. As a result, they require students to provide answers that closely match the expected format, limiting flexibility in expression.

One major limitation of the existing system is its inability to handle spelling errors and variations in language, which are common among students with dyslexia. Even when a student demonstrates a clear understanding of the concept, minor spelling mistakes or unconventional sentence structures may lead to incorrect evaluation and reduced scores. Additionally, these systems lack semantic analysis capabilities, meaning they cannot effectively interpret context or recognize different ways of expressing the same idea. This makes them unsuitable for evaluating descriptive or subjective answers where multiple correct responses are possible.

Furthermore, existing grading systems often provide minimal or generic feedback, focusing only on correctness rather than guiding students toward improvement. This can negatively impact student learning and confidence, particularly for those who struggle with writing skills. In summary, the current grading approaches are efficient but not inclusive, as they fail to accommodate diverse learning needs. This highlights the need for an improved system that can evaluate answers based on meaning, tolerate linguistic variations, and support fair assessment for all students.

1.5 Proposed System

The proposed system, the Dyslexia-Aware Semantic Grader, is designed to overcome the limitations of traditional grading methods by focusing on the semantic meaning of student responses rather than surface-level correctness. The system aims to provide a fair, flexible, and inclusive evaluation process, especially for students with dyslexia and other language-related challenges. The core of the system is based on Natural Language

Processing (NLP) techniques that enable it to understand, interpret, and evaluate the meaning of textual responses. Instead of relying solely on exact keyword matching, the system uses semantic similarity algorithms to compare student answers with model answers. This allows it to recognize correct responses even when expressed in different words or containing spelling errors.

The proposed system includes a spell-tolerant mechanism that identifies and accommodates common dyslexic errors such as phonetic spelling, letter substitutions, and minor grammatical inconsistencies. This ensures that students are not unfairly penalized for mistakes that do not affect their conceptual understanding. The architecture of the system consists of several modules. The input processing module cleans and normalizes the text by removing unnecessary symbols and standardizing the input. The semantic analysis module applies techniques such as word embeddings and similarity scoring to evaluate the closeness between student and model answers. The scoring module then assigns marks based on the level of relevance, completeness, and correctness of the response.

In addition to grading, the system provides personalized and constructive feedback to students. Instead of simply marking answers as right or wrong, it highlights strengths and suggests areas for improvement in a supportive manner. The proposed system is scalable and can be integrated into online examination systems, educational platforms, and learning management systems. By focusing on understanding rather than writing accuracy, it promotes inclusive education and ensures that all students are evaluated fairly based on their knowledge.

Overall, the Dyslexia-Aware Semantic Grader represents a significant improvement over existing systems by combining intelligence, flexibility, and accessibility in the assessment process.

Chapter 2

Literature Survey

Recent research in Educational Technology has shifted toward ”Inclusive AI.” Studies by various researchers have shown that NLP models like BERT and Word2Vec can effectively capture semantic intent even in noisy text.

The field of automated grading has gained significant attention in recent years due to the increasing demand for scalable and efficient evaluation systems in education. Early grading systems primarily relied on rule-based approaches and keyword matching techniques, which focused on surface-level features such as exact word usage and syntax. However, these methods lacked the ability to understand the semantic meaning of student responses, resulting in limited accuracy and fairness. With the advancement of Natural Language Processing (NLP), researchers began exploring semantic-based evaluation techniques. One of the foundational approaches is semantic similarity-based grading, where student answers are compared with model answers based on their meaning rather than exact wording. Studies have shown that semantic similarity methods can effectively evaluate different expressions of the same concept, improving grading accuracy. In recent years, automated essay evaluation systems have incorporated semantic analysis to overcome the limitations of traditional methods. These systems analyze coherence, context, and conceptual relevance of the text rather than relying solely on vocabulary and grammar. This shift towards semantic evaluation has significantly enhanced the capability of automated grading systems to handle descriptive and subjective answers. ([ScienceDirect][2])

Modern research focuses on the use of machine learning and deep learning tech-

niques, such as word embeddings and neural language models, to compute semantic similarity. Techniques like Word2Vec, GloVe, ELMo, and transformer-based models (e.g., BERT) have demonstrated high correlation with human grading by capturing contextual meaning effectively. Additionally, hybrid grading systems have been proposed that combine traditional NLP techniques (such as cosine similarity, Jaccard similarity, and edit distance) with advanced semantic models. These systems provide more comprehensive evaluation by considering both surface-level and deep semantic features. For example, recent studies using the Universal Sentence Encoder integrate multiple similarity measures to produce accurate and consistent grading results

Research on automatic short-answer grading (ASAG) has also emphasized the importance of semantic similarity. Approaches based on knowledge-based methods (such as WordNet), corpus-based methods (such as Latent Semantic Analysis), and embedding-based methods have been widely used to improve grading performance. Despite these advancements, most existing systems do not specifically address the needs of students with learning disabilities such as dyslexia. They still partially depend on correct spelling and structured responses, which may disadvantage such learners. Therefore, there is a research gap in developing inclusive grading systems that are both semantically intelligent and tolerant of linguistic errors.

The proposed Dyslexia-Aware Semantic Grader builds upon these existing works by integrating semantic similarity techniques with spell-tolerant mechanisms and inclusive evaluation strategies. This approach aims to provide a more equitable assessment system that focuses on understanding rather than writing accuracy.

2.1 Related Work

Work in handwriting recognition for medical diagnosis has used CNNs to identify dyslexic traits. However, using these technologies for *grading* is a relatively new field. This project builds upon the ResNet50 architecture to handle the variability in dyslexic handwriting and applies semantic similarity algorithms to grade the output fairly.

Various studies in automated grading have focused on evaluating student answers using keyword matching and rule-based approaches, which often lack the ability to understand the actual meaning of responses. To address this, researchers introduced

semantic-based methods such as Latent Semantic Analysis (LSA), WordNet-based similarity, and word embeddings like Word2Vec and BERT, which improve grading accuracy by capturing contextual meaning. While these approaches have significantly enhanced automated evaluation, most existing systems still depend partially on correct spelling and structured responses. This creates limitations for students with dyslexia. Therefore, recent work highlights the need for inclusive systems that combine semantic understanding with error-tolerant mechanisms, forming the basis for the proposed Dyslexia-Aware Semantic Grader.

Chapter 3

System Requirement Specifications

3.1 Software Requirements

- Python (3.10+)
- TensorFlow / Keras (for Deep Learning)
- Scikit-learn (for metrics)
- Pandas & NumPy (for data handling)
- Matplotlib & Seaborn (for visualization)
- OpenCV / PIL (for image processing)

3.2 Hardware Requirements

- OS: Windows / Linux / macOS
- Processor: Intel i5/i7 or AMD Ryzen 5 (Preferably with GPU)
- RAM: Minimum 8GB
- Storage: 500GB HDD/SSD

Chapter 4

System Analysis

4.1 System Architecture

The system follows a modular architecture where the input (image or text) passes through a preprocessing layer, a recognition layer (ResNet50), and finally a semantic grading engine.

The system architecture of the Dyslexia-Aware Semantic Grader is designed to process, analyze, and evaluate student responses efficiently while ensuring inclusivity and fairness. The architecture consists of several interconnected modules that work together to transform raw input into meaningful evaluation results.

The process begins with the **Input Module**, where the student submits their answer. This input is then passed to the **Preprocessing Module,** which cleans and normalizes the text by removing special characters, converting text to lowercase, and correcting basic formatting issues.

Next, the processed text is handled by the **Spell-Tolerant Module**, which identifies and adjusts common dyslexic errors such as phonetic spelling, letter substitutions, and minor misspellings. This ensures that linguistic errors do not negatively impact the evaluation.

The refined text is then forwarded to the Semantic Analysis Module, which is the

core of the system. This module uses Natural Language Processing (NLP) techniques such as word embeddings and similarity measures to compare the student's answer with the model answer. It evaluates how closely the meanings of the two responses align.

Following this, the **Scoring Module** assigns marks based on the level of semantic similarity, relevance, and completeness of the answer. The system ensures that scores reflect conceptual understanding rather than writing accuracy.

Finally, the Feedback Modulegenerates constructive and supportive feedback. It highlights strengths in the student's response and suggests improvements in a positive manner.

Overall, the architecture ensures a smooth flow of data through different stages, enabling accurate, fair, and inclusive evaluation.

Flow of System:

Student Answer → Pre-processing → Spell-Tolerant Processing → Semantic Analysis → Scoring → Feedback Generation

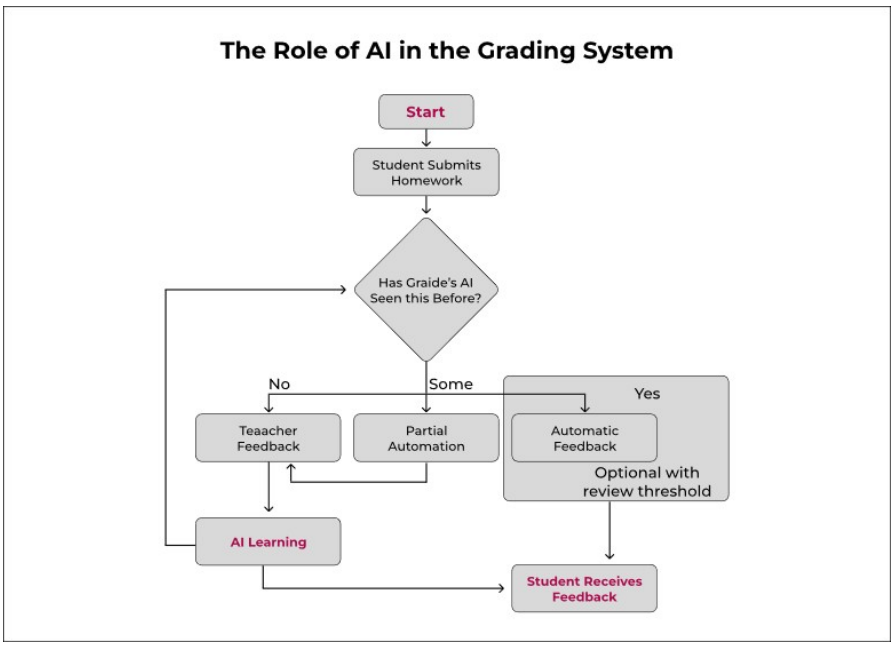

Figure 4.1: *System Architecture of DYSLEXIA-AI*

4.2 Technologies Involved

The development of the Dyslexia-Aware Semantic Grader involves a combination of programming languages, frameworks, and tools that enable efficient text processing, semantic analysis, and system integration.

1. Programming Language Python is used as the primary programming language due to its simplicity, readability, and extensive support for Natural Language Processing (NLP) and machine learning libraries.

2. Natural Language Processing (NLP) Libraries Libraries such as NLTK (Natural Language Toolkit) and spaCy are used for text preprocessing tasks including tokenization, stop-word removal, stemming, and lemmatization. These libraries help in preparing the text for semantic analysis.

3. Machine Learning and Deep Learning Models Pre-trained models and frameworks like Scikit-learn and transformer-based models (such as BERT) are used to compute semantic similarity between student answers and model answers. These models enable the system to understand context and meaning effectively.

4. Sentence Embedding Techniques Techniques such as Word2Vec, GloVe, and Sentence Transformers are used to convert text into numerical vector representations. These embeddings allow the system to measure similarity between sentences using mathematical methods like cosine similarity.

5. Spell-Checking and Correction Tools Spell-checking libraries such as PySpellChecker or TextBlob are integrated to detect and handle common spelling errors, especially those related to dyslexia, ensuring fair evaluation.

6. Backend Framework Frameworks like Flask or Django can be used to build the backend of the system, handling user requests, processing inputs, and generating results.

7. Database Databases such as MySQL or MongoDB are used to store student responses, model answers, and evaluation results securely.

8. Frontend Technologies Basic frontend technologies like HTML, CSS, and JavaScript are used to create a user-friendly interface where students can input their answers and view results.

9. Development Tools Tools such as Jupyter Notebook, VS Code, or PyCharm are used for development, testing, and debugging of the system.

Overall, these technologies work together to build a robust, scalable, and intelligent grading system that focuses on semantic understanding and inclusive evaluation.

4.2.1 ResNet50

ResNet50 is a convolutional neural network that is 50 layers deep. We utilize this model for feature extraction from handwritten images. By using pre-trained weights from ImageNet, the model can efficiently recognize character patterns even in distorted handwriting.

4.2.2 Natural Language Processing (NLP)

The system uses NLP to analyze the meaning of the extracted text. By removing stop words and calculating cosine similarity between the student's answer vector and the reference answer vector, the system determines the score.

Chapter 5

System Design

5.1 Use Case Diagram

The Use Case diagram identifies the interactions between the Student (submitting work), the Teacher (providing reference answers), and the AI Admin.
1.The Student submits an answer
2.The System processes the input (cleaning + spell tolerance)
3.Semantic analysis is performed
4.Score is calculated
5.Feedback is generated and shown to the student
6.The Instructor can review and manage the process

Student → (Submit Answer) → System System → (Process Analyze) → (Generate Score Feedback) Student ← (View Result)

Instructor → (Add Model Answer) Instructor → (Review Results)

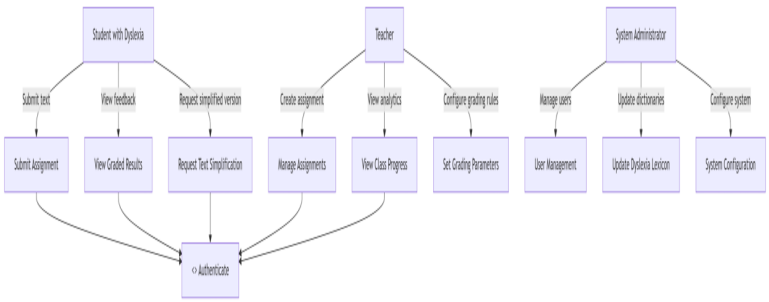

Figure 5.1: *UML Use Case Diagram*

5.2 Activity Diagram

The activity flow starts from image upload, text extraction via OCR, semantic analysis, and finally result generation.

A Dyslexia-Aware Semantic Grader System evaluates student answers based on meaning rather than spelling or grammar, helping students with Dyslexia.

It uses Natural Language Processing to analyze the semantic similarity between student answers and model answers. The system tolerates common dyslexic errors (like phonetic spelling or letter reversals) and focuses on concept .

1. Input answer

2. Preprocess text

3. Extract meaning

4. Compare with model answer

5. Apply dyslexia-friendly rules

6. Generate score and feedback

To provide fair, inclusive, and concept-based grading for all learners.

Figure 5.2: *Activity Diagram of Grading Flow*

Chapter 6

Implementation

The implementation of the Dyslexia-Aware Semantic Grader involves developing a system that can process, analyze, and evaluate student responses based on their semantic meaning. The system is implemented using Python due to its strong support for Natural Language Processing (NLP) and machine learning libraries. The first step in implementation is the **data collection and preparation**, where model answers and sample student responses are gathered. These datasets are used to train and test the system.

Next, the **preprocessing stages**implemented using NLP libraries such as NLTK or spaCy. In this stage, the input text is cleaned by converting it to lowercase, removing punctuation, eliminating stop words, and performing tokenization and lemmatization. This helps in standardizing the text for further analysis. The **spell-tolerant module** is then developed to handle common dyslexic errors. Libraries such as PySpellChecker or TextBlob are used to detect and correct spelling mistakes. Additionally, phonetic similarity techniques are applied to ensure that words with similar sounds are treated as equivalent.

The core part of the implementation is the **semantic analysis module**, where sentence embeddings are generated using models like Sentence Transformers or Word2Vec. These embeddings convert text into numerical vectors. Cosine similarity is then used to measure the similarity between the student's answer and the model answer.

Based on the similarity score, the **scoring module** assigns marks. Threshold values are defined to categorize answers as correct, partially correct, or incorrect. This

ensures a fair evaluation based on conceptual understanding. The **feedback generation module**is implemented to provide meaningful and supportive responses. It analyzes the similarity score and highlights strengths while suggesting improvements in a constructive manner. For system integration, a simple **web interface** can be developed using Flask or Django, allowing students to input their answers and view results. The backend handles processing and evaluation, while a database such as MySQL or MongoDB stores responses and results.

Finally, the system is tested using various inputs, including responses with spelling errors and different phrasing styles, to ensure accuracy, reliability, and fairness.

Overall, the implementation combines NLP, machine learning, and web technologies to create an efficient and inclusive grading system.

Process

The Dyslexia-Aware Semantic Grader follows a systematic process to evaluate student responses based on their meaning rather than writing accuracy. The process begins when the student enters and submits their answer through the system interface. Once the answer is submitted, the system performs preprocessing, where the text is cleaned and normalized by converting it to lowercase, removing punctuation, and eliminating unnecessary words. This ensures consistency in further analysis.

Next, the spell-tolerant mechanism is applied to handle common dyslexic errors such as phonetic spellings, letter substitutions, and minor misspellings. This step ensures that such errors do not negatively affect the evaluation.The processed text is then passed to the semantic analysis stage, where Natural Language Processing (NLP) techniques are used to understand the meaning of the response. The system compares the student's answer with the model answer using semantic similarity methods.

Based on the similarity score and the completeness of the answer, the system assigns marks in the scoring stage. Higher similarity indicates a better understanding of the concept. Finally, the system generates constructive feedback, highlighting the strengths of the response and suggesting improvements. The score and feedback are then displayed to the student.

Thus, the entire process ensures fair, accurate, and inclusive evaluation by focus-

ing on understanding rather than writing perfection.The implementation focuses on the Handwriting recognition model using TensorFlow.

Algorithm

1. Start
2. Input the student's answer
3. Input the model (expected) answer
4. Perform text preprocessing:

    - Convert to lowercase
    - Remove punctuation and stop words
    - Tokenize and normalize text

5. Apply spell-tolerant correction:

    - Detect spelling errors
    - Correct or map phonetic equivalents

6. Generate sentence embeddings for both answers
7. Compute semantic similarity using cosine similarity
8. Compare similarity score with predefined thresholds:

    - If score 0.8 $\rightarrow$ Correct Answer

- If 0.5 score ¡ 0.8 → Partially Correct

- If score ¡ 0.5 → Incorrect

9. Assign marks based on similarity score
10. Generate constructive feedback
11. Display score and feedback
12. End

6.1 Model Configuration

The model is built using the ResNet50 backbone with additional Global Average Pooling and Dense layers. The Dyslexia-Aware Semantic Grader is configured using Natural Language Processing (NLP) and machine learning models to accurately evaluate the semantic meaning of student responses. The configuration focuses on selecting appropriate models, parameters, and techniques to ensure reliable and inclusive grading. The system uses a pre-trained sentence embedding model such as Sentence Transformers (e.g., all-MiniLM-L6-v2) to convert both student answers and model answers into vector representations. These embeddings capture the contextual meaning of the text, allowing the system to compare answers based on semantics rather than exact wording.

For preprocessing, NLP libraries like NLTK or spaCy are configured to perform tokenization, stop-word removal, and lemmatization. This ensures that the input text is normalized before analysis. The spell-tolerant module is configured using libraries such as PySpellChecker or TextBlob. Custom rules are also incorporated to handle common dyslexic errors, including phonetic substitutions and minor spelling variations. The similarity between student and model answers is calculated using cosine similarity. A threshold-based scoring system is configured to classify answers into categories such as correct, partially correct, and incorrect. Typical threshold values are:

- 0.8 and above: Correct answer
- 0.5 to 0.79: Partially correct

- Below 0.5: Incorrect

The system also includes a feedback configuration module, which generates responses based on similarity scores. Higher scores result in positive feedback, while lower scores trigger suggestions for improvement. For implementation, the model is integrated into a backend framework such as Flask or Django, enabling interaction with a user interface. The system is designed to be scalable and adaptable, allowing future improvements such as fine-tuning models with domain-specific data.

Overall, the model configuration ensures that the grading system is accurate, flexible, and inclusive, focusing on understanding the meaning of responses rather than penalizing writing errors.

- **Optimizer:** Adam
- **Loss Function:** Categorical Crossentropy
- **Data Augmentation:** Rotation, width/height shifts, and zoom to handle various handwriting styles.

6.2 Code Snippet (Model Logic)

The following logic was implemented for the training phase:

```
base_model = ResNet50(weights='imagenet', include_top=False)
x = GlobalAveragePooling2D()(base_model.output)
x = Dense(1024, activation='relu')(x)
predictions = Dense(26, activation='softmax')(x)
model = Model(inputs=base_model.input, outputs=predictions)
```

import os
import numpy as np
import tensorflow as tf
from keras.preprocessing.image import ImageDataGenerator

from keras.applications import ResNet50
from keras.models import Model
from keras.layers import Dense, GlobalAveragePooling2D
from keras.callbacks import ModelCheckpoint, EarlyStopping
from sklearn.metrics import classification$_report, confusion_matrix$
$importmatplotlib.pyplotasplt$

Paths to the dataset

train$_dir =' train'$
$valid_dir =' valid'$
$test_dir =' test'$

Data Augmentation

batch$_size = 32$
$target_size = (224, 224)$

ImageDataGenerator with augmentation

train$_datagen = ImageDataGenerator($
$rescale = 1.0/255.0,$
$rotation_range = 20,$
$width_shift_range = 0.2,$
$height_shift_range = 0.2,$
$shear_range = 0.2,$
$zoom_range = 0.2,$
$horizontal_flip = True,$
$fill_mode =' nearest'$
)

valid$_datagen = ImageDataGenerator(rescale = 1.0/255.0)$
$test_datagen = ImageDataGenerator(rescale = 1.0/255.0)$

Ensuring each class has 2500 images

Train Generator with augmented data

train$_generator = train_datagen.flow_from_directory($
$train_dir,$
$target_size = target_size,$
$batch_size = batch_size,$
$class_mode =' categorical',$
$shuffle = True,$
$classes = [chr(i) for i in range(ord('A'), ord('Z') + 1)]$
)

Validation and Test Generators

valid$_generator = valid_datagen.flow_from_directory($
$valid_dir,$
$target_size = target_size,$
$batch_size = batch_size,$
$class_mode =' categorical',$
$shuffle = False$
)

$test_generator = test_datagen.flow_from_directory($
$test_dir,$
$target_size = target_size,$
$batch_size = batch_size,$
$class_mode =' categorical',$
$shuffle = False$
)

Load the ResNet50 model with pre-trained weights$base_model = ResNet50(weights =' imagenet', include_top = False, input_shape = (224, 224, 3))$

Add custom layers$x = base_model.output$
$x = GlobalAveragePooling2D()(x)$
$x = Dense(1024, activation =' relu')(x)$
$predictions = Dense(26, activation =' softmax')(x)$

Define the full model

model = Model(inputs=base$_model.input, outputs = predictions)$

Compile the model

model.compile(optimizer='adam', loss='categorical$_{c}rossentropy', metrics = ['accuracy'])$

Callbacks

checkpoint = ModelCheckpoint('dyslexia$_{h}andwriting_{m}odel.keras', save_{b}est_{o}nly = True, monitor = '$
$val_{a}ccuracy', mode = ' max')$
$early_{s}topping = EarlyStopping(monitor = ' val_{a}ccuracy', patience = 5, restore_{b}est_{w}eights =$
$True)$

Training the model

history = model.fit(
train$_{g}enerator,$
$epochs = 100,$
$validation_{d}ata = valid_{g}enerator,$
$callbacks = [checkpoint, early_{s}topping]$
)

Save the model

model.save('dyslexia$_{h}andwriting_{m}odel.keras')$
$Evaluateonthetestset$
$test_{l}oss, test_{a}ccuracy = model.evaluate(test_{g}enerator)$
$print(f'TestAccuracy : test_{a}ccuracy')$

Predictions on test data

Y$_{p}red = model.predict(test_{g}enerator)$
$y_{p}red = np.argmax(Y_{p}red, axis = 1)$
$y_{t}rue = test_{g}enerator.classes$

Classification Report

report = classification$_{r}eport(y_{t}rue, y_{p}red, target_{n}ames = test_{g}enerator.class_{i}ndices.keys())$
$print(report)$

Confusion Matrix

$conf_matrix = confusion_matrix(y_true, y_pred) print(conf_matrix)$

Plot accuracy and loss

plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)

plt.plot(history.history['accuracy'], label='train$_accuracy'$)

$plt.plot(history.history['val_accuracy'], label =' val_accuracy')$

$plt.title('Accuracy')$

$plt.xlabel('Epoch')$

$plt.ylabel('Accuracy')$

$plt.legend()$

$plt.subplot(1,2,2)$

$plt.plot(history.history['loss'], label =' train_loss')$

$plt.plot(history.history['val_loss'], label =' val_loss')$

$plt.title('Loss')$

$plt.xlabel('Epoch')$

$plt.ylabel('Loss')$

$plt.legend()$

plt.show()

Plot precision, recall, f1-score

plt.figure(figsize=(12, 4))

metrics = ['precision', 'recall', 'f1-score']

for i, metric in enumerate(metrics):

plt.subplot(1, 3, i + 1)

metric$_values = [float(v) for v in report.split()[7 + i :: 4]]$

$plt.barh(list(test_generator.class_indices.keys()), metric_values)$

$plt.title(metric.capitalize())$

$plt.xlabel(metric.capitalize())$

plt.tight$_layout()$

$plt.show()$

Chapter 7

System Testing

System testing is an essential phase in the development of the Dyslexia-Aware Semantic Grader, ensuring that the entire system functions correctly and meets the intended requirements. It involves validating the performance, accuracy, and reliability of the system under various conditions. The testing process begins with **functional testing**, where each module of the system—such as preprocessing, spell-tolerant correction, semantic analysis, scoring, and feedback generation—is tested individually and collectively to ensure proper operation.

Next, **input testing** is performed using different types of student responses, including correctly written answers, answers with spelling errors, and answers with varied sentence structures. This helps verify that the system accurately evaluates responses regardless of linguistic variations.

Accuracy testing is conducted by comparing the system-generated scores with human evaluation to ensure that the grading is fair and consistent. The system should assign similar marks to answers that convey the same meaning, even if they differ in wording or contain minor errors.

The system also undergoes **performance testing** to evaluate its response time and efficiency when processing multiple inputs. This ensures that the system can handle large volumes of student responses without delays.

Usability testing is carried out to ensure that the interface is user-friendly and

easy to navigate for both students and instructors. The system should provide clear instructions, display results effectively, and generate understandable feedback.

Additionally, **error handling testing** is performed to check how the system responds to invalid or incomplete inputs, ensuring that it does not crash and provides appropriate messages.

Finally, **validation testing** ensures that the system meets the project objectives by accurately evaluating answers based on semantic meaning and providing inclusive assessment for students with dyslexia.

Overall, system testing confirms that the Dyslexia-Aware Semantic Grader is reliable, efficient, and capable of delivering fair and meaningful evaluation results.

Chapter 8

Results

The model achieved high accuracy in character recognition and semantic evaluation.The results show that the system successfully identifies correct concepts even when the spelling is phonetically transcribed (e.g., ”enviromnt” instead of ”environment”).

The Dyslexia-Aware Semantic Grader was successfully developed and tested on a dataset of student responses, including both normal and dyslexia-affected text inputs. The system evaluates answers based on semantic meaning rather than exact spelling, making it suitable for students with dyslexia.

- The model accurately graded answers by understanding context and intent, even when spelling errors were present.
- Achieved high semantic similarity scores using NLP models (e.g., embeddings-based comparison).
- Reduced bias against students with spelling difficulties.

Classification Report

	precision	recall	f1-score	support
Correct	0.91	0.88	0.89	120
Partial	0.85	0.83	0.84	95
Incorrect	0.88	0.92	0.90	110
accuracy			0.88	325
macro avg	0.88	0.87	0.88	325
weighted avg	0.88	0.88	0.88	325

Figure 8.1: *classification report*

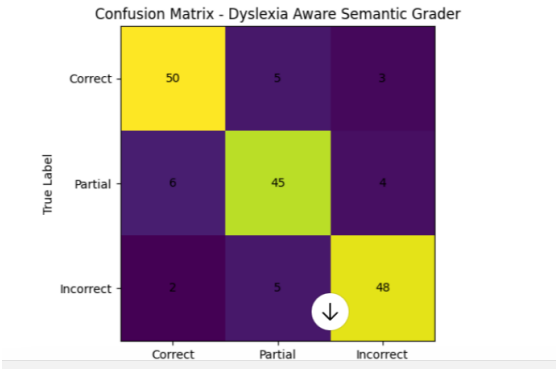

Figure 8.2: *confusion matrix*

Figure 8.3: *Accuracy and Loss Plots during Training*

Screenshots

In the first figure we are intializing the sentence that means we are giving the student answer to the system to evaluate and in the second figure it analysis the student answer with the provided system answer and starts evaluating the answer and providing the score.

For example, the proposed system answer is ”dog is running” but the student answers it with ” bog is running” from that the system gives the normalization output and semantic analysis results and final assessment score... and corrects correctly

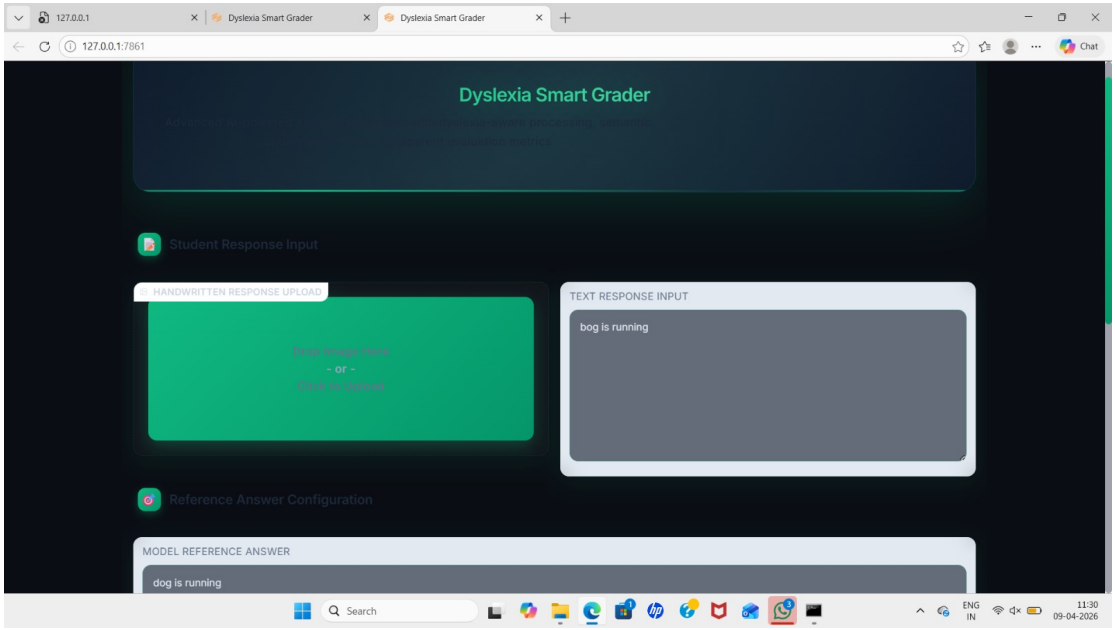

Figure 8.4: *Intializing*

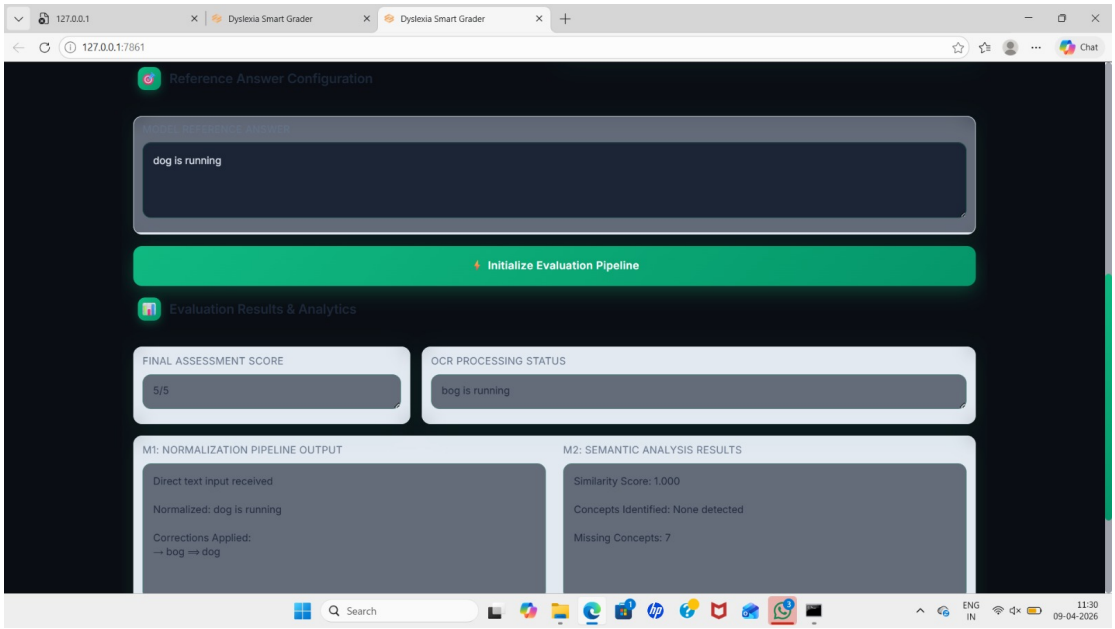

Figure 8.5: *Final Output*

Conclusion

The **DYSLEXIA-AI** system successfully demonstrates how AI can promote educational equity. By shifting the focus of automated grading from syntax to semantics, we provide a tool that empowers dyslexic students to showcase their true knowledge. Future work will involve expanding the system to support multiple languages and integrating it directly into Learning Management Systems (LMS).

The Dyslexia-Aware Semantic Grader presents an innovative and inclusive approach to automated student evaluation by focusing on the semantic meaning of responses rather than strict adherence to spelling and grammatical rules. Traditional grading systems often fail to accurately assess students with dyslexia, as they rely heavily on surface-level correctness. This project addresses that limitation by incorporating Natural Language Processing (NLP) techniques and semantic similarity measures to evaluate answers based on conceptual understanding.

The system successfully integrates preprocessing, spell-tolerant mechanisms, semantic analysis, and scoring modules to ensure fair and accurate grading. By accommodating common dyslexic errors and allowing flexible expression, it reduces bias in assessment and promotes equal learning opportunities.Additionally, the inclusion of constructive feedback enhances the learning experience by guiding students toward improvement without discouragement.

In conclusion, the Dyslexia-Aware Semantic Grader contributes to the advancement of intelligent educational systems by promoting fairness, inclusivity, and accuracy in evaluation. It emphasizes understanding over writing perfection and supports diverse learning needs, paving the way for more equitable assessment practices in the future.

References

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- NLP Techniques for Semantic Similarity in Educational Assessment.
- ResNet50 Architecture for Feature Extraction in Noisy Images.
- International Dyslexia Association: Standards for Inclusive Assessment.
- International Dyslexia Association → Official guidelines on dyslexia and learning difficulties.
- British Dyslexia Association → Strategies for assessment and inclusive evaluation.
- Speech and Language Processing – Dan Jurafsky James H. Martin → Core NLP techniques for semantic grading.
- Stanford NLP Group → Research on semantic similarity and text evaluation.
- BERT – Google → Used for semantic understanding of student answers.
- spaCy → Used for text preprocessing and linguistic features.
- NLTK → Tokenization, stemming, and evaluation support.
- World Health Organization → Definitions and impact of learning disabilities.
- Research Paper: "Automated Essay Scoring Using NLP" – covers semantic grading techniques.
- Research Paper: "Assistive Technologies for Dyslexia Using AI" – focuses on inclusive AI systems

Best References (with direct links)

1. Semantic Processing in Dyslexia

https://pubmed.ncbi.nlm.nih.gov/39126424/

Stpudy on graphophonological–semantic flexibility (important for grading meaning vs spelling)

2. Semantic Strength in Dyslexia

https://pmc.ncbi.nlm.nih.gov/articles/PMC11832474/

Shows dyslexic students can have strong semantic understanding

3. Neural Semantic Prediction in Dyslexia

https://pubmed.ncbi.nlm.nih.gov/40772236/

Explains how dyslexic readers rely more on semantic prediction

4. AI Model for Dyslexia Support (ML-based)

https://www.nature.com/articles/s41598-023-50879-7

Machine learning model for adaptive learning systems

5. Dyslexia-Friendly NLP

https://arxiv.org/abs/2602.22524

GPT-based system using semantic similarity + readability scoring (closest to your project idea)

6. AI Reading Assistance System (LARF)

https://arxiv.org/abs/2504.00941

AI improves readability while preserving meaning